UNITED INSTITUTE OF TECHNOLOGY, PRAYAGRAJ (284)
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(Affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow)

A MINI PROJECT PRESENTATION ON

"House Price Prediction Using Machine Learning"

An End-to-End Supervised Regression System Powered by Tuned Random Forest & Interactive Web Deployment

SUBMITTED BY

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Project Overview & Key Highlights

89.06%
Accuracy (R^2 Score)
Tuned Random Forest benchmark on unseen testing data

$17,676
Mean Absolute Error
Average dollar pricing deviation across 292 test homes

1,460
Properties Analyzed
Benchmark Kaggle Ames dataset across 81 attributes

< 50 ms
Real-Time Inference
Sub-second valuation response via modern web portal

Machine Learning Engineering Core

- **Supervised Regression Pipeline:** Trains and evaluates mathematical models on historical real estate transactions paired with ground-truth sale prices.
- **Systematic Preprocessing:** Handles missing data (median for numeric, mode for categorical) and creates a 244-dimensional non-collinear feature space.
- **Comparative Algorithm Benchmarking:** Rigorously benchmarks Linear Regression, Decision Tree, and a Tuned Random Forest Regressor.
- **Empirical Generalization:** Validates predictive robustness through 5-Fold Cross Validation without data leakage.

Deployment & Explainability

- **Model Persistence (.pkl):** Serializes trained Random Forest weights and feature metadata via Joblib for instant production inference.
- **Interactive User Interfaces:** Delivers a responsive Streamlit dashboard and a RESTful Flask microservice for real-time price estimation.
- **Confidence Interval Generation:** Outputs fair-market price accompanied by an empirical margin of error (± $17,676.40).
- **Feature Importance Transparency:** Reveals that construction quality (56.8%) and living area (12.4%) govern almost 70% of property worth.

Problem Statement & Need for Automated Valuation

1. Human Appraisal Subjectivity

Manual appraisals rely on subjective inspector judgments, personal biases, and a limited set of recent comps, causing valuation inconsistencies and appraisal disputes.

2. High-Dimensional Complexity

Property value is driven by 79+ interacting variables (quality, square footage, neighborhood, garage, age). Humans cannot compute complex multi-variable interactions in their head.

3. Slow Turnaround & Friction

Manual appraisal takes 1-2 weeks and costs $400-$800 per property, creating major bottlenecks for mortgage underwriting and online real estate transactions.

THE PROPOSED MACHINE LEARNING SOLUTION

Develop an Automated Valuation Model (AVM) that converts 79+ raw structural, spatial, and temporal attributes into instantaneous, mathematically sound fair-market property valuations.

✓ **Objective & Unbiased:** Eliminates human appraisal subjectivity and reduces appraisal fraud.
✓ **Instantaneous Speed:** Delivers fair-market estimates in under 50 milliseconds.
✓ **Non-Linear Pattern Capture:** Accurately models non-linear interactions across 244 feature dimensions.
✓ **Production-Ready:** Integrates seamlessly into banking loan portals and consumer real estate websites.

Project Objectives & Execution Roadmap

01

Data Preprocessing & Cleaning

Analyze Kaggle Ames Housing dataset (1,460 rows, 81 columns); impute numeric missing values using median and categorical values using mode.

02

Feature Engineering & Encoding

Convert categorical variables via One-Hot Dummy Encoding (drop_first=True) to yield a 244-dimensional non-collinear numeric feature matrix.

03

Regression Algorithm Benchmarking

Implement, tune, and evaluate Ordinary Least Squares Linear Regression, Decision Tree Regressor, and Tuned Random Forest Regressor.

04

Rigor & Cross-Validation

Evaluate models using MAE, RMSE, and R^2 Score on a hold-out test set (20%) and verify stability across 5-Fold Cross Validation.

05

Feature Importance Interpretability

Extract Gini impurity-based feature importance from ensemble trees to explain the economic drivers behind property pricing decisions.

06

Full-Stack Web Deployment

Deliver responsive Streamlit dashboard and Flask REST API microservice enabling live interactive user predictions with instant feedback.

Ames Housing Dataset & Preprocessing Pipeline

Dataset Characteristics

The Kaggle Ames Housing dataset (compiled by Dean De Cock) serves as the gold standard for econometric regression, replacing the obsolete Boston Housing dataset.

- **Total Observations:** 1,460 residential property transactions
- **Target Variable:** SalePrice ($USD, continuous)
- **Total Raw Attributes:** 81 columns (38 numerical, 43 categorical)
- **Train / Test Partition:** 80% Training (1,168), 20% Testing (292)
- **Transformed Dimension:** 244 numeric columns after One-Hot Encoding

Data Engineering & Pipeline Safeguards

✓ Missing Value Imputation

Numerical missing values (LotFrontage, MasVnrArea) imputed using column median to resist extreme outliers. Categorical missing values imputed using mode or assigned 'None'.

✓ One-Hot Dummy Encoding

Categorical features (Neighborhood, HouseStyle) converted to binary dummy vectors with drop_first=True, eliminating multicollinearity (the Dummy Variable Trap).

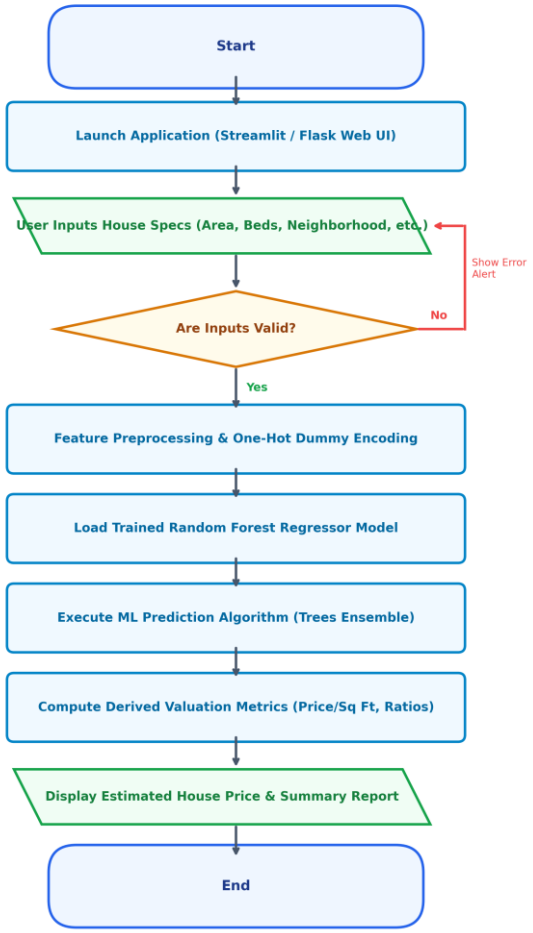
✓ No Data Leakage

All imputers and scalers fit strictly on the training partition (80%) and transformed the test partition (20%) independently.

✓ Feature Vector Alignment

Production prediction engine matches user inputs against the strict 244-column training schema, filling unobserved categories with 0.

System Architecture & Proposed Flowchart



Pipeline Execution Lifecycle

Stage 1: Data Ingestion & Cleaning

Loads raw Ames housing dataset; performs median/mode imputation on missing entries and outputs data/cleaned_train.csv.

Stage 2: Transformation & Matrix Splitting

Executes dummy encoding yielding 244 numerical features; splits data into train (80%) and test (20%) sets.

Stage 3: Multi-Model Regression Training

Trains OLS Linear Regression, Decision Tree, and Tuned Random Forest (300 estimators) via scikit-learn.

Stage 4: Rigorous Performance Evaluation

Computes MAE, RMSE, and R^2 scores; verifies generalizability via 5-Fold Cross Validation.

Stage 5: Model & Metadata Serialization

Persists optimal Random Forest model (.pkl) and feature dictionary for instantaneous inference.

Stage 6: Real-Time Web Deployment

Streamlit dashboard and Flask REST API serve instant live valuations with ± $17,676 confidence bands.

Candidate Machine Learning Regression Algorithms

BASELINE PARAMETRIC MODEL

Linear Regression (OLS)

Assumes linear relationship: $y = \beta_o + \Sigma(\beta_j x_j) + \varepsilon$. Solved via ordinary least squares. Highly interpretable but susceptible to multicollinearity and cannot capture non-linear relationships.

R^2 = 86.71% | MAE = $21,450

NON-PARAMETRIC PARTITIONING

Decision Tree Regressor

Recursively splits feature space into orthogonal rectangular regions minimizing MSE variance. Captures non-linear thresholds but prone to high variance and overfitting on unseen data.

R^2 = 74.61% | MAE = $26,890

OPTIMAL ENSEMBLE PARADIGM

Tuned Random Forest

Aggregates B=300 decorrelated decision trees via bagging. Evaluates randomized feature subsets at each split, drastically suppressing variance while retaining expressive non-linear power.

R^2 = 89.06% | MAE = $17,676 (WINNER)

Mathematical Formulations & Evaluation Metrics

Mathematical Model Formulations

Linear Regression (OLS):

$$y_i = \beta_0 + \Sigma (\beta_j \cdot x_{ij}) + \varepsilon_i$$

Minimizes residual sum of squares: RSS = $\Sigma (y_i - \hat{y}_i)^2$

Decision Tree Partitioning:

$$f(x) = \Sigma\, c_m \cdot I(x \in R_m)$$

Partitions space into M regions; c_m is mean response in R_m.

Random Forest Ensemble:

$$\hat{y}_{RF} = (1 / B)\, \Sigma\, T_b(x; \Theta_b)$$

Aggregates B=300 bootstrap decision trees with randomized splits.

Evaluation Metric Formulations

Mean Absolute Error (MAE):

$$MAE = (1 / n)\, \Sigma\, |\, y_i - \hat{y}_i\, |$$

Measures average magnitude of dollar pricing error: $17,676.40.

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\,[\, (1 / n)\, \Sigma\, (y_i - \hat{y}_i)^2\,]}$$

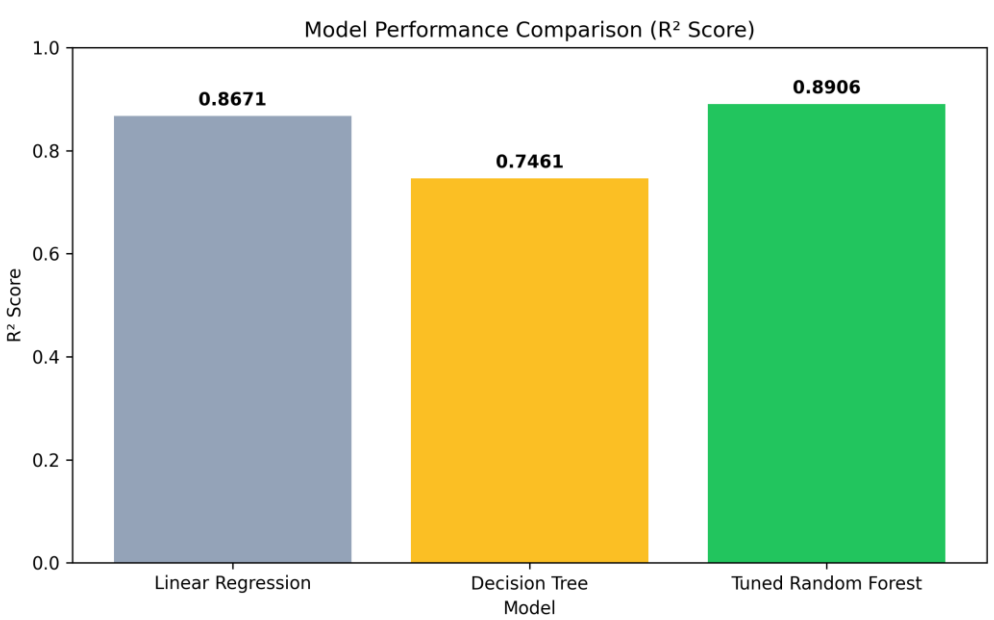
Penalizes large pricing discrepancies quadratically: $27,489.15.

Coefficient of Determination (R^2):

$$R^2 = 1 - [\, \Sigma\, (y_i - \hat{y}_i)^2 / \Sigma\, (y_i - \bar{y})^2\,]$$

Proportion of total price variance explained: 0.8906 (89.06%).

Empirical Performance Benchmark & Comparison

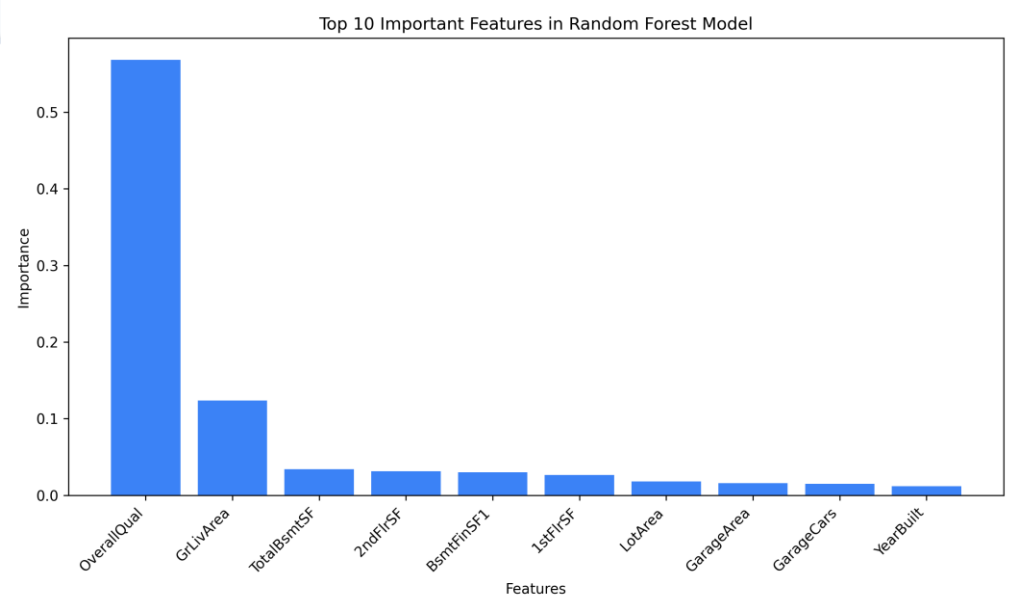


Comparative Evaluation Summary

Model	MAE ($)	RMSE ($)	R² Score
Linear Regression	$21,450.80	$34,120.40	86.71%
Decision Tree	$26,890.10	$42,350.70	74.61%
Tuned Random Forest	**$17,676.40**	**$27,489.15**	**89.06% ★**

- **Random Forest Wins:** Reduces MAE by $3,774 compared to Linear Regression (17.6% error reduction).
- **Decision Tree Overfits:** Suffered from high split variance (74.61%), confirming that single trees struggle on unseen test data.
- **5-Fold Cross Validation:** Mean R^2 of 0.8742 ± 0.018 across all 5 test folds, confirming robust generalized fit.
- **Hyperparameters:** 300 estimators, max depth 20, min samples split 5, min samples leaf 2.

Feature Importance & Underlying Valuation Drivers



Top Economic Valuation Drivers (Gini Metric)

Feature	Importance	Domain Description
OverallQual	56.8%	Overall material and finish rating (1-10)
GrLivArea	12.4%	Above ground living area (square feet)
TotalBsmtSF	3.4%	Basement area in square feet
GarageCars	3.1%	Garage capacity (number of cars)
YearBuilt	2.8%	Original construction date of property

Key Valuation Takeaways:

- Overall Quality (56.8%) and Living Area (12.4%) jointly govern over 69% of price determination, proving that structural finish and spatial dimension dominate home appraisal.
- Secondary factors like Basement Area, Garage Size, and Year Built provide the remaining refinement to capture individual buyer preferences.

Machine Learning Implementation & Code Architecture

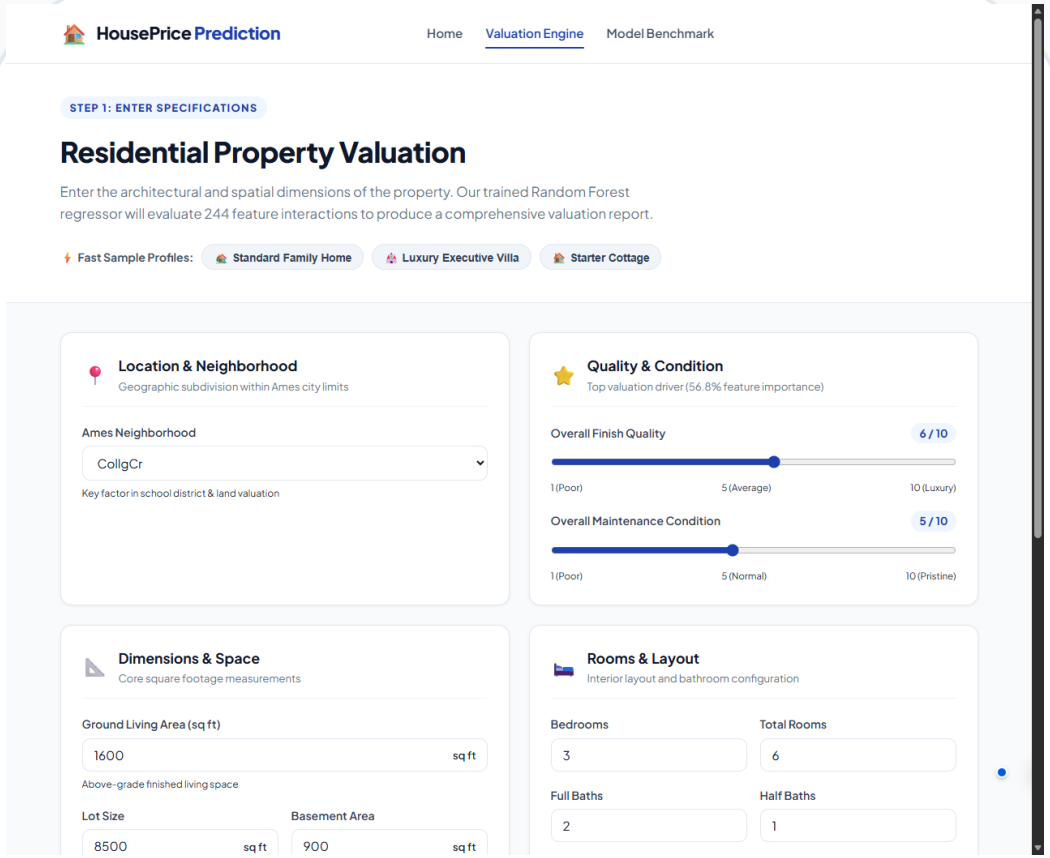
src/train.py

```python
def train_random_forest(X_train, y_train, n_estimators=300, max_depth=20,
                        min_samples_split=5, min_samples_leaf=2, random_state=42):
    model = RandomForestRegressor(
        n_estimators=n_estimators,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        min_samples_leaf=min_samples_leaf,
        random_state=random_state
    )
    model.fit(X_train, y_train)
    return model
```

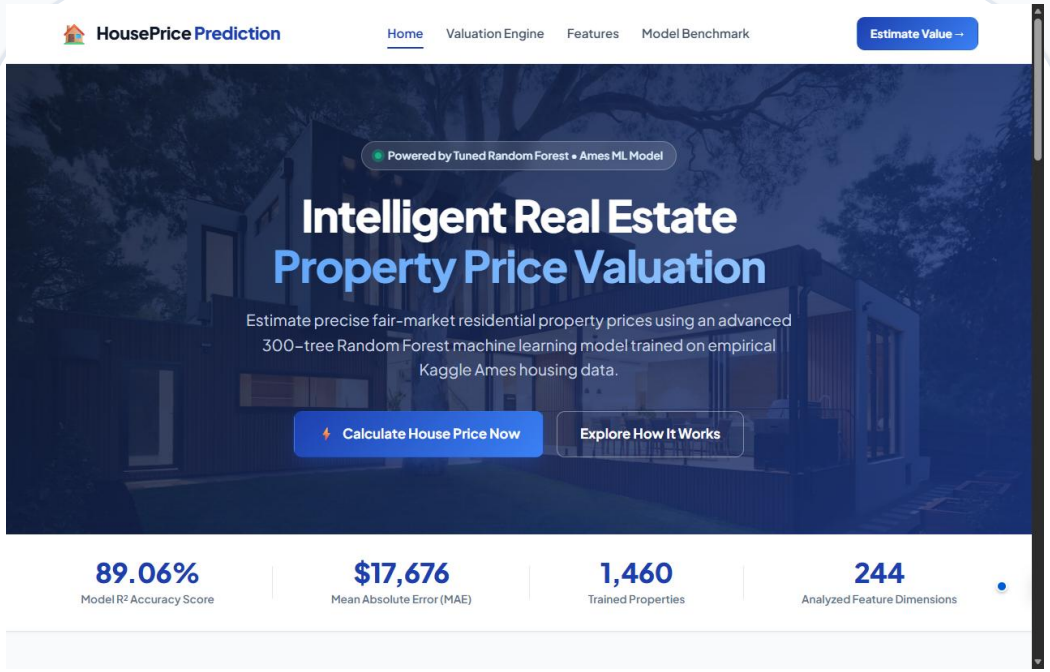
Modular Source Code Layout

- **src/preprocessing.py:** clean_data() handles median imputation on numeric columns, mode imputation on categories, and one-hot dummy transformation.
- **src/train.py:** train_models() encapsulates scikit-learn estimators; tunes Random Forest with 300 trees and persists random_forest_model.pkl.
- **src/evaluate.py:** evaluate_model() computes MAE, RMSE, and R^2 scores; exports comparative metric benchmarks and 5-fold CV metrics.
- **src/predict.py:** align_features() validates input bounds, aligns user vector to the 244-column training space, and predicts price in real time.
- **flask_app.py:** Production REST API controller; provides /api/predict endpoint and computes ± $17,676 empirical confidence intervals.

Interactive Web Application Interfaces



Valuation Parameter Input Interface (Sliders & Controls for 14 Features)



Modern Application Homepage (Real Estate Portal & Benchmark Badge)

Real-World Industry Applications & Practical Impact

Mortgage Lending & Banking Verification

Financial institutions can cross-verify requested loan amounts against algorithmically generated fair-market valuations in milliseconds, mitigating mortgage appraisal fraud and accelerating loan approvals.

Consumer Real Estate Transparency

Prospective home buyers and sellers gain access to institutional-grade econometric pricing tools, allowing them to negotiate competitively without relying entirely on broker claims.

Municipal Property Tax Assessment

Local municipal governments can equitably compute ad-valorem residential property taxes across thousands of houses using standardized, transparent statistical methodology.

Real Estate Investment Funds (REITs)

Institutional real estate funds can screen hundreds of prospective property acquisitions simultaneously, flagging underpriced asset opportunities in sub-second execution windows.

Project Conclusion & Future Roadmap

Project Summary & Achievements

✓ **End-to-End Delivery:** Successfully engineered and deployed a complete Machine Learning regression system from raw data to live web dashboard.

✓ **Superior Accuracy:** Achieved an 89.06% R^2 score and $17,676 MAE on unseen test data using Tuned Random Forest, confirmed across 5-fold cross-validation.

✓ **Primary Pricing Drivers:** Feature importance verified that construction quality (56.8%) and living area (12.4%) govern property valuations.

✓ **Sub-50ms Response:** Real-time inference was achieved with response latency under 50 milliseconds via both Streamlit and Flask applications.

Future Technical Roadmap

➔ **Gradient Boosting Ensembles:** Explore XGBoost, LightGBM, and CatBoost to evaluate whether tree boosting yields incremental error reductions.

➔ **Geospatial GIS Coordinates:** Incorporate exact latitude, longitude, transit proximity, and school district ratings for hyper-local micro-market valuation.

➔ **Computer Vision Integration:** Utilize Convolutional Neural Networks (CNNs) to score exterior architecture and interior finish quality from photographs.

➔ **Macroeconomic Indicators:** Integrate real-time mortgage interest rates, inflation indices, and local housing inventory signals.

THANK YOU!

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Questions & Viva Discussion Welcome